RotateAttention: RoPE-Aware Rotation & Range Rectification

for INT4 Quantized Attention in Video Generation

ECCV 2026 Submission #1552 · Yaofu Liu, Wanli Lan, Jinxi Li, Binhang Yuan, Harry Yang

Core Idea

Fast, accurate **INT4 FlashAttention** for 3D-RoPE video DiTs. 3D RoPE induces *segmental* $\mathbf{Q}/\mathbf{K}$ outliers with cross-matrix symmetry; $\mathbf{P} \in (0, 1]$ wastes *half* of the INT4 range.

Method

(1) RoPE-aware Rotation: block-diagonal orthogonal $\mathbf{R}$ (Interleaved/Half) removes outliers at negligible cost. **(2) Range-optimized P:** $\tilde{\mathbf{P}} \in (-8, 7]$ uses all 16 INT4 levels. **(3) Hadamard V:** merged offline into $\mathbf{W}_v, \mathbf{W}_o$ (zero cost).

Results

Up to **2.2**× kernel and **1.68**× end-to-end speedup with near-FP16 quality. Learned rotations **must be orthogonal**—unconstrained ones amplify quantization error.